

Stock Risk, Return and Portfolio Analysis of 30 Tehran Stock Exchange Stocks (2021–2026)

Executive Summary

This project investigates the risk-return characteristics of 30 stocks listed on the Tehran Stock Exchange (TSE) over the period from March 2021 to May 2026. Using historical market data, monthly returns were calculated and analyzed to evaluate stock performance, systematic risk exposure, diversification opportunities, and portfolio optimization characteristics.

The study applies key concepts from Modern Portfolio Theory (MPT), Capital Asset Pricing Model (CAPM), and risk-adjusted performance measurement to compare investment opportunities across different stocks.

Dataset Description

The dataset consists of daily trading information for 30 publicly traded companies on the Tehran Stock Exchange.

Dataset Characteristics

- Market: Tehran Stock Exchange (TSE)
- Number of Stocks: 30
- Observation Period: March 2021 – May 2026
- Total Observations: 121,096
- Data Frequency: Daily observations aggregated into monthly observations
- Variables:
 - Adjusted Closing Price (AdjClose)
 - Trading Volume
 - Trading Value
 - Daily High and Low Prices
 - Gregorian Date
 - Jalali Date
 - Stock Symbol

The adjusted closing price was used to calculate investment returns because it provides a more consistent measure of shareholder value over time.

Financial Methodology

1. Monthly Return Analysis

Monthly stock returns were calculated using percentage changes in adjusted closing prices:

$$R_t = \frac{P_t - P_{t-1}}{P_{t-1}}$$

where:

- R_t = Monthly Return
- P_t = Current Month Adjusted Price
- P_{t-1} = Previous Month Adjusted Price

Monthly returns provide a standardized measure of performance across all stocks.

2. Cumulative Return Analysis

To evaluate long-term investment performance, cumulative returns were computed over the entire sample period.

Cumulative return measures the total growth of an investment and captures the compounding effect of returns through time.

This metric was used to identify:

- Best-performing stock over the five-year period
 - Worst-performing stock over the five-year period
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3. Risk Measurement

Investment risk was measured using the standard deviation of monthly returns:

$$\sigma = \sqrt{\frac{\sum (R_i - \bar{R})^2}{n - 1}}$$

Higher standard deviation indicates greater volatility and uncertainty in future returns.

This measure serves as the primary risk indicator in Modern Portfolio Theory.

4. Risk-Return Tradeoff

A risk-return scatter plot was constructed to investigate whether stocks with higher risk generated higher expected returns.

According to financial theory, investors require higher expected returns as compensation for bearing additional risk.

The relationship between:

- Expected Return
- Return Volatility

was examined for all 30 stocks.

5. Correlation Analysis

A correlation matrix was constructed using monthly returns.

Correlation analysis is important because portfolio diversification depends on combining assets with imperfect correlations.

Lower correlations among assets can reduce portfolio risk without necessarily reducing expected returns.

6. Sharpe Ratio Analysis

Risk-adjusted performance was evaluated using the Sharpe Ratio:

$$Sharpe = \frac{R_p - R_f}{\sigma_p}$$

where:

- R_p = Portfolio (or stock) return
- R_f = Risk-free rate
- σ_p = Return volatility

The Sharpe Ratio measures the amount of excess return generated for each unit of risk undertaken.

Stocks with higher Sharpe Ratios provide superior risk-adjusted performance.

7. Efficient Frontier Construction

Portfolio simulations were conducted to generate an Efficient Frontier.

The Efficient Frontier represents the set of portfolios that maximize expected return for a given level of risk.

This analysis illustrates the benefits of diversification and portfolio optimization.

8. Beta Estimation and CAPM Interpretation

Systematic risk was measured using Beta:

$$\beta_i = \frac{Cov(R_i, R_m)}{Var(R_m)}$$

where:

- R_i = Stock Return
- R_m = Market Return

Beta measures a stock's sensitivity to overall market movements.

Interpretation:

- $\beta > 1$: Aggressive Stocks
- $\beta \approx 1$: Market-like Behavior
- $\beta < 1$: Defensive Stocks

This classification helps investors understand market exposure and systematic risk.

Key Findings

Long-Term Performance

The analysis identified both the highest-performing and lowest-performing stocks during the sample period based on cumulative returns.

Risk and Return Characteristics

The results indicate substantial variation in risk and expected return across the 30 analyzed stocks.

Several high-return stocks were associated with elevated levels of volatility, illustrating the fundamental risk-return tradeoff.

Diversification Benefits

Correlation analysis demonstrated that combining stocks with different return patterns can significantly reduce portfolio risk.

Risk-Adjusted Performance

Sharpe Ratio rankings revealed that the highest-return stock is not necessarily the most attractive investment after adjusting for risk.

Systematic Risk

Beta estimates identified both aggressive and defensive stocks, providing insight into how different securities respond to market-wide fluctuations.

Technologies Used

- Python
 - Pandas
 - NumPy
 - Matplotlib
 - Seaborn
 - Jupyter Notebook
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Conclusion

This project applies core concepts of investment management, portfolio theory, and financial risk analysis to a five-year dataset of 30 Tehran Stock Exchange stocks.

The findings demonstrate the practical application of return measurement, volatility analysis, diversification, Sharpe Ratio evaluation, Efficient Frontier construction, and CAPM beta estimation. Together, these techniques provide a comprehensive framework for evaluating investment opportunities and constructing portfolios that balance risk and return.